Entanglement-Entropy Sourcing of Gravity: Dimensional Repair, Covariant Completion, and Experimental Outlook

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Abstract

We repair a dimensional inconsistency in the entanglement–gravity framework and present a refined, self-consistent phenomenological model. The original coupling prefactor $c^4/(8\pi G k_B \ln 2)$ multiplying an entropy density in bits per cubic meter does not yield an energy density; the missing ingredient is a reference energy scale E_* converting bit counts to joules. We introduce the minimal external-source effective action with coupling $\Lambda = \tilde{\kappa} E_*$, vary with respect to the metric only, and derive a dimensionally correct entanglement stress–energy tensor, exact fifth-force law, and $1/R^2$ exterior acceleration scaling for bounded sources. All experimental observables reduce to the product $\Lambda_{\text{eff}} = \tilde{\kappa} E_* \alpha_{\text{screen}}$; neither $\tilde{\kappa}$ nor E_* is individually constrained by measurement. We identify and address a critical constraint from macroscopic superconductors, which requires the operational definition of internal entanglement entropy to exclude thermally mixed correlations. Under the Planck-scale hypothesis $E_* \approx E_P$, near-field force sensing with cryogenic SQUID-readout mechanics is the preferred experimental direction, with a predicted unsuppressed signal of $3.6 \times 10^{-9}\,\mathrm{m/s^2}$ for 100 entangled bits at $100\,\mu$m separation.

1 Purpose and Scope

This document is a dimensional-consistency revision of the entanglement–gravity program. Its goal is not to select a unique microscopic origin for the coupling, but to isolate the minimal scaling assumptions required for a well-posed phenomenological theory.

Earlier drafts used a coupling prefactor proportional to $c^4/(8\pi G k_B \ln 2)$ multiplying an entanglement entropy density measured in bits per unit volume. In SI units, that expression does not yield an energy density, so the resulting stress–energy tensor and derived accelerations were not dimensionally valid. The present revision fixes that problem by introducing a reference energy scale E_* and rebuilding the source term around it.

2 Operational Definition and Units

2.1 The dimensional problem

The Einstein equations

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{1}$$

require $[T_{\mu\nu}] = \mathrm{J\,m^{-3}}$. Writing $[S_{\text{ent}}] = \mathrm{bit\,m^{-3}}$ (bit dimensionless) and using the original coupling $\alpha = \tilde{\kappa} c^4/(8\pi G k_B \ln 2)$ gives $[\alpha S_{\text{ent}}] = \mathrm{K\,m^{-4}}$, not an energy density. A reference energy scale E_* (J/bit) is the minimal repair.

2.2 Protocol-certified entanglement: S_{cert}

The earlier notation S_{cert} was ambiguous: it could be interpreted as the von Neumann entropy of any reduced density matrix, including thermal and BCS correlations in the device substrate. We replace it with a precisely defined operational quantity.

Definition 2.1 (Protocol-certified entanglement density)**.** Let D be a quantum device, P a specified gate sequence acting on the device degrees of freedom, $D = A \cup B$ a bipartition pre-registered before P is applied, and $\mathcal{W}$ an entanglement witness verified after P. The **protocol-certified entanglement density** is

$$S_{\text{cert}}(x) = \frac{-k_B \operatorname{Tr}_A\left[\rho_A^P(x) \ln \rho_A^P(x)\right]}{k_B \ln 2\, \Delta V}, \quad \rho_A^P = \operatorname{Tr}_B\left[P \rho_D P^\dagger\right], \tag{2}$$

where $P\rho_D P^\dagger$ is the device state after protocol P and ΔV is the cell volume. We require that $\mathcal{W}$ certifies $S_{\text{cert}} > 0$ to within experimental error before a run is accepted.

This replaces the generic bipartite von Neumann entropy used in earlier drafts. The definition has three operationally necessary properties: (i) $S_{\text{cert}} \geq 0$; (ii) $S_{\text{cert}} = 0$ if no entangling protocol was applied; (iii) S_{cert} is certified by a positive entanglement witness, so it cannot be mimicked by classical correlations.

3 Minimal Dimensional Fix

3.1 Reference energy scale

We introduce a reference energy scale E_* with units of energy per bit and define the entanglement energy density by

$$\rho_{\text{ent}}(x) = E_*\, S_{\text{ent}}(x). \tag{3}$$

Then

$$[\rho_{\text{ent}}] = [E_*][S_{\text{ent}}] = \mathrm{J\,m^{-3}}, \tag{4}$$

as required.

An equivalent parameterization is

$$\rho_{\text{ent}}(x) = \lambda_* k_B T_* s_{\text{ent}}(x), \tag{5}$$

where s_{ent} is entropy density in thermodynamic units, T_* is an effective temperature scale, and λ_* is dimensionless. In this white paper we use the simpler single-scale notation E_*.

3.2 Revised source term

The entanglement contribution to the stress–energy tensor is rewritten as

$$T_{\mu\nu}^{\text{ent}}(x) = \tilde{\kappa}\, \rho_{\text{ent}}(x) g_{\mu\nu} + (\text{gradient terms}), \tag{6}$$

with

$$\rho_{\text{ent}}(x) = E_* S_{\text{ent}}(x). \tag{7}$$

Thus all previous appearances of αS_{ent} are replaced by $\tilde{\kappa} E_* S_{\text{ent}}$.

If desired, one may later relate E_* to a microscopic scale such as

$$E_* = \eta k_B T_*, \tag{8}$$

with η dimensionless and T_* an effective Unruh or horizon temperature. The present framework leaves the origin of E_* open.

4 Revised Effective Action and Fifth-Force Equation

4.1 External-source action

Let

$$\phi(x) \equiv S_{\text{ent}}(x). \tag{9}$$

We retain the external-source interpretation: ϕ is prescribed by the device quantum state and is not varied as an independent field. The entanglement-sector Lagrangian density is written as

$$\mathcal{L}_{\text{ent}} = -\frac{\beta}{2} g^{\mu\nu}(\nabla_\mu \phi)(\nabla_\nu \phi) - \lambda\phi, \tag{10}$$

with

$$\lambda = \tilde{\kappa} E_*. \tag{11}$$

The full action is therefore

$$S[g;\phi] = \int d^4x \sqrt{-g} \left[\frac{c^4}{16\pi G} R - \frac{\beta}{2} g^{\mu\nu}(\nabla_\mu \phi)(\nabla_\nu \phi) - \tilde{\kappa} E_* \phi \right]. \tag{12}$$

Varying with respect to $g_{\mu\nu}$ yields

$$T^{\text{ent}}_{\mu\nu} = \beta \left(\nabla_\mu \phi \, \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} (\nabla\phi)^2 \right) + \tilde{\kappa} E_* \phi \, g_{\mu\nu}. \tag{13}$$

The dimensions are correct provided β carries the dimensions needed to make the gradient term an energy density. In slowly varying configurations, the gradient term is subleading.

4.2 Matter nonconservation

Total conservation requires

$$\nabla^\mu T^{\text{total}}_{\mu\nu} = 0, \tag{14}$$

so that

$$\nabla^\mu T^{\text{matter}}_{\mu\nu} = -\nabla^\mu T^{\text{ent}}_{\mu\nu}. \tag{15}$$

Substituting the revised entanglement tensor gives

$$\nabla^\mu T^{\text{matter}}_{\mu\nu} = -\left(\tilde{\kappa} E_* + \beta \Box \phi\right) \nabla_\nu \phi + \beta \nabla_\nu \left(\frac{1}{2} (\nabla\phi)^2 \right). \tag{16}$$

In the quasi-static regime with modest gradients and $|\beta \Box \phi| \ll |\tilde{\kappa} E_*|$, this reduces to

$$\nabla^\mu T^{\text{matter}}_{\mu\nu} \approx -\tilde{\kappa} E_* \, \nabla_\nu \phi(x) = -\tilde{\kappa} E_* \, \nabla_\nu S_{\text{ent}}(x). \tag{17}$$

This is the corrected fifth-force law. Its units are now those of force density, as required.

5 Newtonian Limit and Revised Observable

5.1 Modified Poisson equation

In the Newtonian limit,

$$\nabla^2\Phi = 4\pi G\left(\rho_{\rm matter} + \rho_{\rm ent}\right), \tag{18}$$

with

$$\rho_{\rm ent}(\mathbf{x}) = \tilde{\kappa} E_* S_{\rm ent}(\mathbf{x}). \tag{19}$$

Under spherical symmetry,

$$\Delta a(R) = -\frac{d\Phi_{\rm ent}}{dR} = \frac{G}{R^2} M_{\rm ent}(< R), \tag{20}$$

where

$$M_{\rm ent}(< R) = 4\pi \int_0^R \rho_{\rm ent}(r) r^2\, dr. \tag{21}$$

For bounded sources, $M_{\rm ent}(< R)$ saturates outside the source and the correction scales as $1/R^2$.

5.2 Observable combination

Experiments constrain the product

$$\Lambda_{\rm eff} \equiv \tilde{\kappa}\, E_*\, \alpha_{\rm screen}, \quad [\Lambda_{\rm eff}] = \mathrm{J/bit}, \tag{22}$$

not $\tilde{\kappa}$ or E_* individually. In branch-comparison experiments the useful observational combination is

$$\mathcal{B} \equiv |\Lambda_{\rm eff}|\, \frac{\Delta S_{\rm ent}^{\rm tot}}{R_{\rm eff}^2}, \quad [\mathcal{B}] = \mathrm{J\,m^{-3}}, \tag{23}$$

where $\Delta S_{\rm ent}^{\rm tot}$ is the differential internal entanglement content between experimental branches and $R_{\rm eff}$ is the characteristic geometric scale. A null result at sensitivity $\Delta a_{\rm max}$ constrains

$$|\Lambda_{\rm eff}| < \frac{\Delta a_{\rm max}\, R_{\rm eff}^2}{G\, \Delta S_{\rm ent}^{\rm tot}}. \tag{24}$$

Mapping this bound to $|\tilde{\kappa}|$ alone requires an independent assumption about E_*; without such an assumption, only $|\Lambda_{\rm eff}|$ is constrained.

6 Implications for OP8 and OP10

6.1 Fifth-force bounds

The corrected acceleration outside a bounded source of radius R_0 is

$$\Delta a(R) = \frac{G\, |\Lambda_{\rm eff}|\, \Delta S_{\rm ent}^{\rm tot}}{R^2}, \quad R > R_0. \tag{25}$$

A null force bound $\Delta a_{\rm max}$ constrains the observable product

$$|\Lambda_{\rm eff}| = |\tilde{\kappa}|\, E_*\, \alpha_{\rm screen} \lesssim \frac{\Delta a_{\rm max}\, R_{\rm eff}^2}{G\, \Delta S_{\rm ent}^{\rm tot}}. \tag{26}$$

Benchmark (unsuppressed signal). For the Planck-scale hypothesis $E_* = E_P = 1.96 \times 10^9\,\mathrm{J}$ and $|\tilde{\kappa}| = 1/4$, the effective mass per bit is $m_{\mathrm{bit}} = |\tilde{\kappa}| E_P/c^2 = 5.44\,\mu\mathrm{g/bit}$. For $\Delta S_{\mathrm{ent}} = 100\,\mathrm{bits}$ at $R = 100\,\mu\mathrm{m}$ with $\alpha_{\mathrm{screen}} = 1$:

$$\Delta a = \frac{G \times 100 \times 5.44 \times 10^{-9}\,\mathrm{kg}}{(10^{-4}\,\mathrm{m})^2} = 3.6 \times 10^{-9}\,\mathrm{m/s^2}. \tag{27}$$

Cryogenic SQUID-based force sensors achieve $\sim 10^{-10}\,\mathrm{m/s^2}/\sqrt{\mathrm{Hz}}$, placing this signal above the noise floor by a factor of $\sim$36 at 1 Hz bandwidth.

Critical constraint. If $E_* \approx E_P$, macroscopic superconductors containing $\sim 10^{20}$ Cooper pairs would carry an effective entanglement mass of order 10^{11} kg per gram of material, producing enormous gravitational anomalies not observed. The framework requires that the operational definition of S_{cert} excludes uncontrolled thermal correlations (including BCS Cooper-pair entanglement) for which no controlled bipartition protocol was applied. Verifying this claim rigorously is an open problem; see Section 7.

6.2 RG interpretation

With E_* explicit, the natural renormalization-group ansatz becomes

$$\tilde{\kappa}_{\mathrm{lab}} E_{*,\mathrm{lab}} = \tilde{\kappa}_{\mathrm{UV}} E_{*,\mathrm{UV}} \left(\frac{E_{\mathrm{lab}}}{E_P}\right)^{\gamma}, \tag{28}$$

where E_P is the Planck energy and γ is an effective scaling exponent. Experiments therefore constrain the infrared product $\tilde{\kappa} E_*$ rather than $\tilde{\kappa}$ alone.

7 Open Problems

7.1 Cancellation theorem for equilibrium backgrounds

The protocol-certified definition (Section 2.2) resolves the Cooper-pair constraint through a cancellation theorem, not through the incorrect claim that BCS states carry no entanglement. BCS correlations are real; the point is that they are *branch-independent* and drop out of the differential observable.

Proposition 7.1 (Background cancellation)**.** *Let the total system be $T = D \cup B_{\mathrm{g}}$, where D is the quantum device and B_{g} is the background (substrate, Cooper pairs). Assume:*

(i) Product initial state: $\rho_T^{(0)} = \rho_D^{(0)} \otimes \rho_{B_{\mathrm{g}}}^{(0)}$.
(ii) Protocol acts on D only: $P^{A,B} = U_D^{A,B} \otimes \mathbf{I}_{B_{\mathrm{g}}}$.
(iii) Bipartition $\mathcal{A}$ pre-registered on D only.

Then $\Delta S_{\mathrm{cert}}^{B_{\mathrm{g}}} = 0$ exactly.

Proof. Under (ii), $P^{A,B}$ cannot change $\rho_{B_{\mathrm{g}}}$, so the background state is identical in both branches. Under (iii), the bipartition traces only over D. Therefore $S_{\mathrm{cert}}(\rho_{B_{\mathrm{g}}}; P^A) = S_{\mathrm{cert}}(\rho_{B_{\mathrm{g}}}; P^B)$ and the difference vanishes. □

Accordingly, branch-comparison experiments measure only

$$\Delta S_{\mathrm{cert}}^{\mathrm{tot}} = S_{\mathrm{cert}}(\rho_D^A) - S_{\mathrm{cert}}(\rho_D^B), \tag{29}$$

in which the substrate contribution is absent by construction regardless of whether Cooper-pair or other background entanglement exists.

Quantitative example. For 10^{20} Cooper pairs per gram at $E_* = E_P$, the background entanglement energy density is enormous in absolute terms but contributes zero to $\Delta M_{\rm ent}$. The measured signal is $\Delta M_{\rm ent} = E_* \, \Delta S^D_{\rm cert} = E_* \times N_{\rm ebits}$ where $N_{\rm ebits} \leq N_{\rm qubits}$.

Validity of assumption (i). Device-substrate coupling is phononic in transmon systems ($J_{\rm ph} \lesssim 1\,\text{kHz}$ vs. $\omega/2\pi \sim 5\,\text{GHz}$, ratio $\sim 10^{-7}$), so initial entanglement between device and substrate is negligible. Gate-induced phonon backaction is a systematic that must be matched between branches; it does not constitute $S_{\rm cert}$ under the protocol definition.

7.2 Status of E_*

E_* is a matching coefficient that cannot be determined from the framework itself. It parameterizes the unknown UV physics connecting entanglement counting to energy. Three physically motivated choices span many orders of magnitude:

Choice	E_*	Δa (100 bits, 100 μm, $\alpha_{\rm screen} = 1$)
Planck scale ($E_* = E_P$)	1.96×10^9 J	3.6×10^{-9} m/s^2
Qubit energy ($E_* = \hbar\omega$, 5 GHz)	3.3×10^{-24} J	6×10^{-42} m/s^2
Thermal ($E_* = k_B T$, 15 mK)	2.1×10^{-25} J	4×10^{-43} m/s^2

Only the Planck-scale choice produces a signal above any conceivable near-term experimental threshold. The framework is therefore experimentally interesting only if $E_* \gtrsim 10^5$ J, which in turn requires resolving the Cooper-pair constraint via OP-new.

7.3 OP3: Lindblad screening numerics

A staged numerical program determines $\alpha_{\rm screen}(N, \tau)$ for realistic NISQ platforms: exact Liouvillian evolution for $N \leq 10$; Kraus-mapped gate errors for $N \leq 20$; MPO-TDVP tensor-network methods for $N \approx 50$. The key diagnostic is whether $\alpha_{\rm screen}(50, T_\phi) \gtrsim 10^{-2}$ for cluster-state initial conditions. Analytical predictions (confirmed for $N = 10$) give $\alpha_{\rm screen} \approx e^{-\gamma_\phi \tau}$ for cluster states (independent of N) vs. $e^{-N\gamma_\phi \tau}$ for GHZ states, strongly motivating cluster-state protocols.

7.4 OP10: Planck-scale renormalization

Whether $\tilde{\kappa} = -1/4$ at the Planck scale renormalizes to an observable laboratory value depends on the scaling exponent γ in $\tilde{\kappa}_{\rm lab} E_{*,\rm lab} = \tilde{\kappa}_{\rm UV} E_{*,\rm UV} (E_{\rm lab}/E_P)^\gamma$. For $\gamma = 0$ (marginal), no renormalization occurs. For $\gamma = 4$, the laboratory coupling is suppressed by $(E_{\rm lab}/E_P)^4 \approx 10^{-132}$. Power counting and the classical-limit requirement constrain γ; this problem is accessible to standard EFT techniques.

8 Programmatic Consequences

The dimensional repair does not alter the operational definition of $S_{\rm ent}(x)$, the Lindblad screening program, or the computation of $\alpha_{\rm screen}(N, \tau)$. What changes is the mapping from entanglement content to physical source strength and the correct form of the observable combination: $\Lambda_{\rm eff} = \tilde{\kappa} E_* \alpha_{\rm screen}$ rather than $\tilde{\kappa} \alpha_{\rm screen}$ alone.

For the Planck-scale hypothesis $E_* = E_P$, near-field force sensing is the preferred experimental direction: the acceleration signal ($3.6 \times 10^{-9}\,\mathrm{m/s^2}$) exceeds the sensitivity of cryogenic SQUID-based force sensors, while Ramsey-interferometry phase shifts are suppressed by c^2 and lie below current thresholds by many orders of magnitude.

9 Conclusions

The entanglement–gravity framework is now dimensionally consistent: introducing E_* repairs the units, and the resulting effective action, stress–energy tensor, matter nonconservation law, and observable combination all carry correct SI dimensions.

Three points define the current state of the framework:

1. $\Lambda_{\mathrm{eff}} = \tilde{\kappa} E_* \alpha_{\mathrm{screen}}$ **is the only constrained combination.** Experiments bound this product; separating $\tilde{\kappa}$, E_*, and α_{screen} requires additional theoretical input (OP10, OP3, and the Cooper-pair argument respectively).

2. **The Cooper-pair constraint is the critical open problem.** If $E_* \approx E_P$ (the only experimentally viable hypothesis), the framework requires that uncontrolled thermal correlations contribute zero to S_{cert} under the operational definition. This must be proven, not assumed.

3. **Force sensing is the correct experimental direction.** For $E_* = E_P$, $|\tilde{\kappa}| = 1/4$, $\Delta S_{\mathrm{ent}} = 100\,\mathrm{bits}$, $R = 100\,\mu\mathrm{m}$, the predicted acceleration is $\Delta a = 3.6 \times 10^{-9}\,\mathrm{m/s^2}$, within reach of cryogenic SQUID-based force sensors. Ramsey interferometry is suppressed by c^2 and is not the primary experimental path.

With these three points explicit, the framework is well-posed, falsifiable, and ready for theoretical and experimental development.